

# NEHA SHARMA - U24AI050 LAB-3

03 February 2026 22:13

## 1. Write a Shell Script to find,

### a) Largest of N numbers

nano largestnumber.sh

```
-----  
#nano side code  
echo "Enter number of elements:"  
read n  
  
echo "Enter numbers:"  
read -a arr  
  
largest=${arr[0]}  
  
for ((i=1; i<n; i++))  
do  
    if [ ${arr[i]} -gt $largest ]  
    then  
        largest=${arr[i]}  
    fi  
done  
  
echo "Largest number is: $largest"
```

```
-----  
chmod +x largestnumber.sh  
./largestnumber.sh
```

### b) Swap two values

nano swapTwoNumber.sh

```
-----  
echo "Enter first number:"  
read a  
echo "Enter second number:"
```

```
read b
```

```
temp=$a
```

```
a=$b
```

```
b=$temp
```

```
echo "After swapping:"
```

```
echo "a = $a"
```

```
echo "b = $b"
```

---

```
chmod +x swapTwoNumber.sh
```

```
./swapTwoNumber
```

### c) Odd or Even

```
nano evenOdd.sh
```

---

```
echo "Enter a number:"
```

```
read num
```

```
if [  $$(num \% 2)$  -eq 0 ]
```

```
then
```

```
    echo "Even number"
```

```
else
```

```
    echo "Odd number"
```

```
fi
```

---

```
chmod +x evenOdd.sh
```

```
./evenOdd.sh
```

### d) Sum of three numbers

```
nano sumOf3.sh
```

---

```
echo "Enter three numbers:"
```

```
read a b c
```

```
sum=$((a + b + c))
```

```
echo "Sum = $sum"
```

chmod +x sumOf3.sh  
./sumOf3.sh

```
ubuntu@ubuntu:~$ nano largestnumber.sh
ubuntu@ubuntu:~$ chmod +x largestnumber.sh
ubuntu@ubuntu:~$ ./largestnumber.sh
Enter number of elements:
4
Enter numbers:
9
./largestnumber.sh: line 12: [: -gt: unary operator expected
./largestnumber.sh: line 12: [: -gt: unary operator expected
./largestnumber.sh: line 12: [: -gt: unary operator expected
Largest number is: 9
ubuntu@ubuntu:~$ nano swapTwoNumber.sh
ubuntu@ubuntu:~$ ./swapTwoNumber.sh
bash: ./swapTwoNumber.sh: Permission denied
ubuntu@ubuntu:~$ chmod +x swapTwoNumber.sh
chmod: cannot access 'swapTwoNumber.sh': No such file or directory
ubuntu@ubuntu:~$ chmod +x swapTwoNumber.sh
ubuntu@ubuntu:~$ ./swapTwoNumber.sh
Enter first number:
5
Enter second number:
8
After swapping:
a = 8
b = 5
ubuntu@ubuntu:~$
```

```
ubuntu@ubuntu:~$ nano evenOdd.sh
ubuntu@ubuntu:~$ chmod +x evenOdd.sh
ubuntu@ubuntu:~$ ./evenOdd.sh
Enter a number:
3
Odd number
ubuntu@ubuntu:~$ nano SumOf3.sh
ubuntu@ubuntu:~$ chmod +x SumOf3.sh
ubuntu@ubuntu:~$ ./SumOf3.sh
Enter three numbers:
3 4 6
Sum = 13
ubuntu@ubuntu:~$
```

2. Write a shell script to check to see if the file “file\_path” exists. If it does exist, display “file\_path”

passwords are enabled.” Next, check to see if you can write to the file. If you can, display “You have permissions to edit “file\_path.”” If you cannot, display “You do NOT have permissions to edit “file\_path”.

```
nano filePath.sh
```

```
-----  
echo "Enter file path:"  
read file_path  
  
if [ -e "$file_path" ]  
then  
    echo "$file_path passwords are enabled."  
  
    if [ -w "$file_path" ]  
    then  
        echo "You have permissions to edit $file_path."  
    else  
        echo "You do NOT have permissions to edit $file_path."  
    fi  
else  
    echo "File does not exist."  
fi
```

```
-----  
chmod +x filePath.sh  
./filePath.sh
```

3. Write a shell script that asks the user to type a number of words as input to shell script Store the input words by user into an array. Print how long each word is.

```
nano print_length.sh
```

```
-----  
echo "Enter words separated by space:"  
read -a words  
  
for word in "${words[@]}"  
do  
    echo "Word: $word , Length: ${#word}"  
done  
-----
```

```
chmod +x print_length.sh
./print_length.sh
```

#### 4. Write a shell program to reverse the five-digit integer.

```
nano reverseFiveDigit.sh
```

```
-----
echo "Enter a five-digit number:"
read num
```

```
rev=0
```

```
while [ $num -gt 0 ]
do
    digit=$((num % 10))
    rev=$((rev * 10 + digit))
    num=$((num / 10))
done
```

```
echo "Reversed number: $rev"
```

```
-----
chmod +x reverseFiveDigit.sh
./reverseFiveDigit.sh
```

#### 5. Write a shell program to search for a given number from the list of numbers using binary search.

```
nano binary_search.sh
```

```
-----
echo "Enter sorted numbers:"
read -a arr
```

```
echo "Enter number to search:"
read key
```

```
low=0
high=$(( ${#arr[@]} - 1 ))
found=0
```

```
while [ $low -le $high ]
do
```

```
mid=$((low + high) / 2))
```

```
if [ ${arr[mid]} -eq $key ]
```

```
then
```

```
    echo "Element found at position $((mid + 1))"
```

```
    found=1
```

```
    break
```

```
elif [ ${arr[mid]} -lt $key ]
```

```
then
```

```
    low=$((mid + 1))
```

```
else
```

```
    high=$((mid - 1))
```

```
fi
```

```
done
```

```
if [ $found -eq 0 ]
```

```
then
```

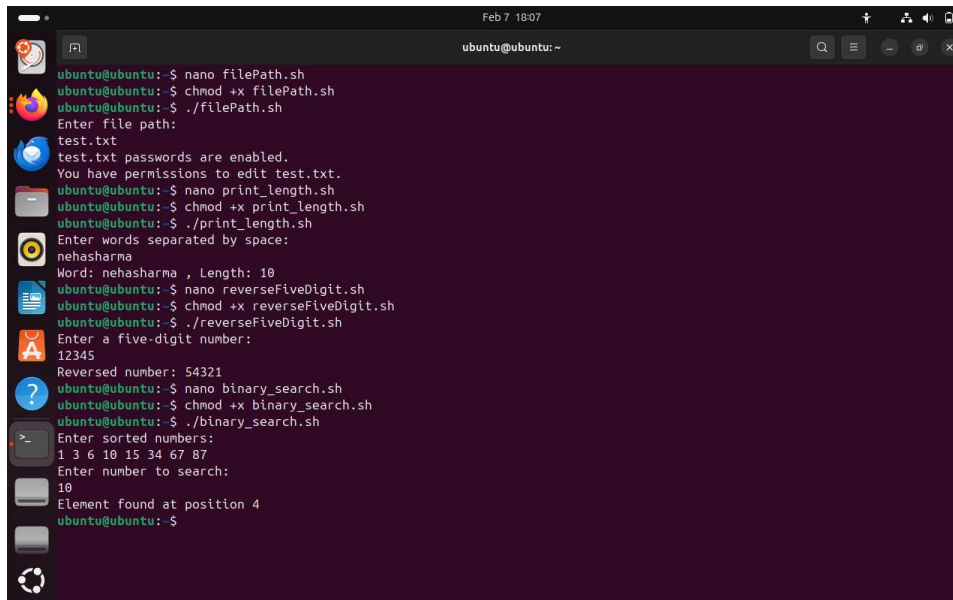
```
    echo "Element not found"
```

```
fi
```

---

```
chmod +x binary_search.sh
```

```
./binary_search.sh
```



```
Feb 7 18:07
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ nano filePath.sh
ubuntu@ubuntu:~$ chmod +x filePath.sh
ubuntu@ubuntu:~$ ./filePath.sh
Enter file path:
test.txt
test.txt passwords are enabled.
You have permissions to edit test.txt.
ubuntu@ubuntu:~$ nano print_length.sh
ubuntu@ubuntu:~$ chmod +x print_length.sh
ubuntu@ubuntu:~$ ./print_length.sh
Enter words separated by space:
nehasharma
Word: nehasharma , Length: 10
ubuntu@ubuntu:~$ nano reverseFiveDigit.sh
ubuntu@ubuntu:~$ chmod +x reverseFiveDigit.sh
ubuntu@ubuntu:~$ ./reverseFiveDigit.sh
Enter a five-digit number:
12345
Reversed number: 54321
ubuntu@ubuntu:~$ nano binary_search.sh
ubuntu@ubuntu:~$ chmod +x binary_search.sh
ubuntu@ubuntu:~$ ./binary_search.sh
Enter sorted numbers:
1 3 6 10 15 34 67 87
Enter number to search:
10
Element found at position 4
ubuntu@ubuntu:~$
```

